Technical Specification

Riverside Community Clinic, Block B (fictional practice project)

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Section 01 00 00 - General Requirements

01.1 The Contractor shall execute the Works in accordance with this Specification, the Drawings and relevant national standards.

01.2 The Contractor shall plan the Works so as to achieve completion within 15 months from the date of commencement, including all finishing and external works.

01.3 All materials shall be new and of approved make. Samples shall be submitted to the Engineer for approval before bulk procurement.

Section 02 30 00 - Earthwork

02.1 Excavation for foundations shall be carried out to the lines and levels shown on the Drawings in all types of soil.

02.2 Backfilling shall be done in layers not exceeding 200 mm, each layer compacted to 95% of maximum dry density.

Section 03 30 00 - Cast-in-Place Concrete

03.1 Concrete grades: Plain cement concrete (PCC) below footings shall be grade C15. Reinforced concrete in footings shall be grade C30. Columns, beams and slabs shall be grade C30.

03.2 Minimum clear cover to reinforcement: footings 50 mm; columns 40 mm; beams 25 mm; slabs 20 mm.

03.3 Concrete shall be design-mix, produced in an approved batching plant. Cube tests shall be carried out at the rate of one set of 3 cubes per 30 m3 or part thereof.

03.4 Curing shall be continued for a minimum of 7 days for all structural concrete.

Section 03 20 00 - Concrete Reinforcement

03.5 Reinforcement steel shall be high-strength deformed bars of grade Fe500D conforming to the relevant national standard.

03.6 Laps shall be staggered and shall not exceed 50% of bars at any one section.

Section 04 20 00 - Unit Masonry

04.1 External walls shall be 230 mm thick solid concrete block masonry in cement mortar 1:6. Internal partitions shall be 115 mm thick in cement mortar 1:4.

Section 07 10 00 - Waterproofing

07.1 Roof waterproofing shall be an APP-modified bituminous membrane of minimum 4 mm thickness, laid over a primed screed and lapped 100 mm at joints.

07.2 The membrane shall be tested by flooding with water for 48 hours before protection screed is laid.

Section 09 30 00 - Tiling

09.1 Floor tiles in clinical areas shall be 600 x 600 mm vitrified tiles, anti-skid, laid on 20 mm cement mortar bed.